

BHARANEE B

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SUMMARY

Computer Science graduate from VIT Chennai, concurrently pursuing the B.S. in Data Science & Applications at IIT Madras, with hands-on experience across backend engineering, computer vision, and applied machine learning. Led a 3-member capstone team to build a real-time multimodal driver-risk assessment system that outperformed published research benchmarks and shipped a production-style full-stack platform with secure authentication, caching, and async processing. Comfortable owning a problem end-to-end — from data pipeline to deployable system.

TECHNICAL SKILLS

Languages: Python, Java, C/C++, SQL, JavaScript, HTML/CSS

Backend & Web: Flask, FastAPI, REST APIs, Vue.js, JWT & RBAC, Celery, Redis, SQLAlchemy

AI / ML: PyTorch, TensorFlow/Keras, Scikit-learn, Hugging Face Transformers, OpenCV, YOLOv8, XGBoost, LightGBM, CatBoost, Pandas, NumPy

Tools & Core CS: Git/GitHub, Docker, Postman, Data Structures & Algorithms, OOP, DBMS

EDUCATION

Vellore Institute of Technology Chennai

2022 – 2026

B.Tech. Computer Science Engineering • CGPA: 7.93

Indian Institute of Technology Madras

2022 – Present

B.S. Data Science and Applications (currently at Diploma level) • CGPA: 6.49

PROJECTS

Driver Risk Assessment System (DRAS) — Final-Year Capstone — Team of 3

Tech Stack: Python, PyTorch, OpenCV, YOLOv8, ResNet-34, Scikit-learn

- Engineered an edge-deployable multimodal pipeline fusing facial-expression recognition (ResNet-34), adaptive eye-aspect-ratio fatigue detection, head-pose estimation, and YOLOv8-based phone-distraction detection into one real-time driver risk score.
- Designed a late-fusion scoring model (70:30 visual-behavioral to affective weighting) across 4 independent perception streams, achieving **93.17% accuracy** on the NTHU-DDD drowsiness benchmark — beating the published 90.48% baseline and 3 alternative classifiers (MLP, KNN, SVM).
- Built continuous baseline-calibration logic to remove false positives from fixed-threshold fatigue detection, enabling real-time inference on standard consumer hardware with no per-driver training data.

Placement Portal Application — Full-Stack Campus Recruitment Platform | [GitHub](#)

Tech Stack: Python, Flask, Vue.js, Redis, Celery, JWT, SQLAlchemy

- Built a decoupled REST API (Flask Blueprints) with JWT-secured Role-Based Access Control powering distinct Admin, Company, and Student workflows across the full campus placement lifecycle.
- Introduced Redis-backed caching and Celery-driven asynchronous background jobs (e.g. CSV exports) to cut database load and eliminate blocking on high-traffic endpoints such as the job board.
- Shipped a responsive Vue.js 3 / Vite single-page app with Progressive Web App support for native-like installation across devices.

Applied Machine Learning — Kaggle Competitions

Tech Stack: Python, PyTorch, Hugging Face Transformers, XGBoost, LightGBM, CatBoost

- Built an end-to-end PyTorch semantic-segmentation pipeline (U-Net/DeepLab-style, Dice loss, IoU evaluation) to classify 6 industrial hardware components at pixel level.
- Benchmarked Linear Regression, Random Forest, XGBoost, LightGBM, and CatBoost for an industrial equipment price-prediction challenge, with full EDA, feature engineering, and cross-validated hyperparameter tuning.
- Fine-tuned Hugging Face transformer models for multiple-choice question answering, optimizing top-3 ranking via the MAP@3 metric.

CERTIFICATIONS

- [Diploma in Programming](#) — IIT Madras
- [Advanced Certificate in Programming and Application Development](#) — IIT Madras
- [Foundational Level in Programming and Data Science](#) — IIT Madras